EE352 Automatic Control

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Frequency Domain Analysis

1. Bode diagram analysis
2. Nyquist analysis
3. Root-locus analysis

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Frequency domain analysis

- If $\lim_{t\to\infty} u(t)$ exists and $G(s)$ is stable, then

$$\lim_{t\to\infty} y(t) = G(0) \lim_{t\to\infty} u(t)$$

- Above result may be generalised to the case where $u(t)$ is a sinusoidal input $\lim_{t\to\infty} [u(t) - \sin \omega t] = 0$

$$\lim_{t\to\infty} \left[y(t) - |G(j\omega)| \sin\left(\omega t + \angle G(j\omega)\right)\right] = 0$$

- Where $|G(j\omega)|$ is the **<u>gain</u>** of $G(s)$ and $\angle G(j\omega)$ is the **<u>phase</u>** of $G(s)$ at frequency ω

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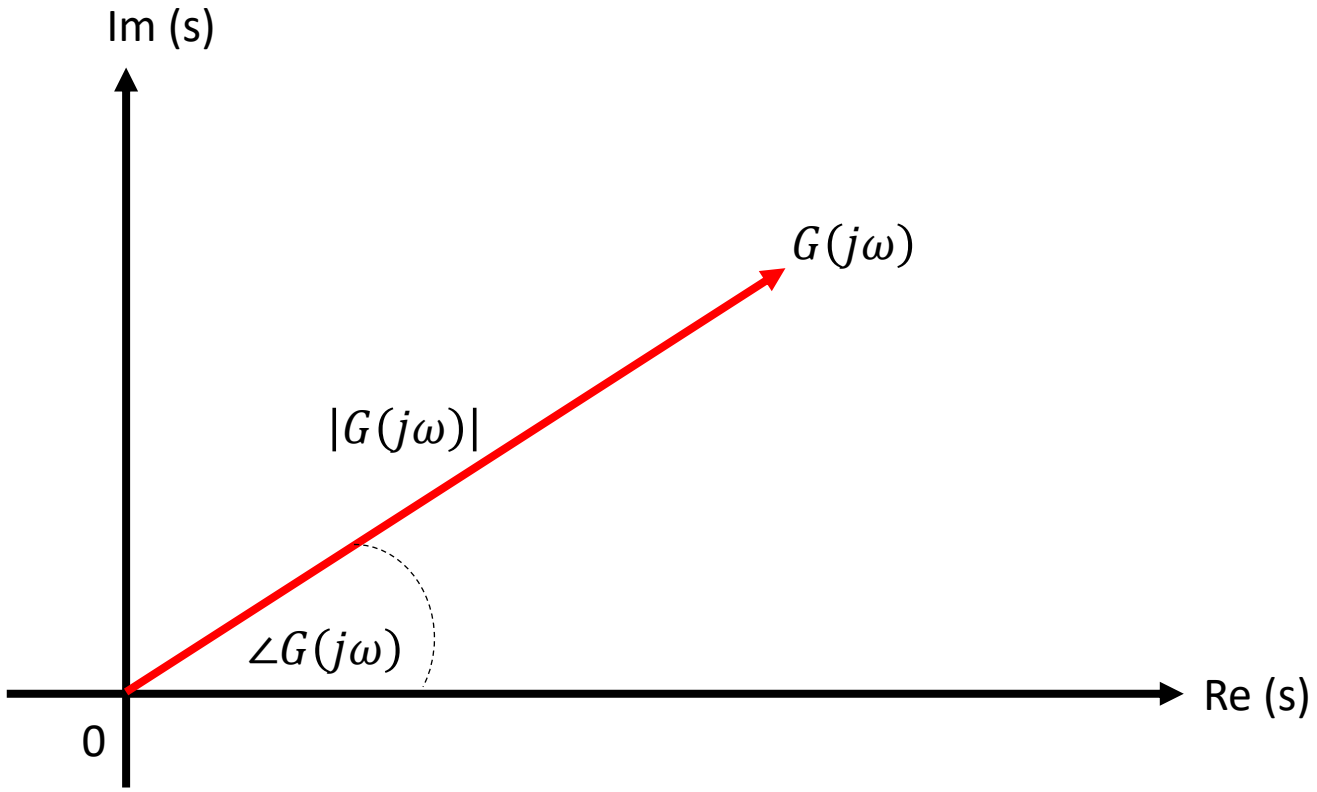
Frequency domain analysis contd.,



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Example 1: $G(s) = \frac{k}{s+a}$

- $G(j\omega) = \frac{k}{j\omega+a}$,
- $|G(j\omega)| = \frac{k}{\sqrt{\omega^2+a^2}}$,
- $\angle G(j\omega) = \tan^{-1}\left(\frac{-\omega}{a}\right) = -\tan^{-1}\left(\frac{\omega}{a}\right)$
- When ω is small, $|G(j\omega)| \approx \frac{k}{a}$ and $\angle G(j\omega) \approx 0$
- When ω is large, $|G(j\omega)| \approx \frac{k}{\omega}$ and $\angle G(j\omega) \approx -\frac{\pi}{2}$

Bode plot

- The Bode plot shows gain and phase against frequency on separate graphs.
- Suppose $G(j\omega) = \frac{G_1(j\omega)G_2(j\omega)}{G_3(j\omega)}$,
- If $|G_i(j\omega)| = g_i, \angle G_i(j\omega) = \phi_i, i = \{1 \quad 2 \quad 3\}$, then
- $|G(j\omega)| = \frac{g_1 g_2}{g_3}, \angle G_i(j\omega) = \phi_1 + \phi_2 - \phi_3$
- The $|G(j\omega)|$ is usually calculated in dB because of easy manipulations.

$$|G(j\omega)|dB = 20\log_{10}|G(j\omega)|$$

- For the above case, $|G(j\omega)|dB = 20\log_{10} g_1 + 20\log_{10} g_2 - 20\log_{10} g_3$

Example 1 contd.,: $G(s) = \frac{k}{s+a}$

- $G(j\omega) = \frac{k}{j\omega+a}, |G(j\omega)| = \frac{k}{\sqrt{\omega^2+a^2}}, \angle G(j\omega) = \tan^{-1}\left(\frac{-\omega}{a}\right) = -\tan^{-1}\left(\frac{\omega}{a}\right)$

- When ω is small, $|G(j\omega)| \approx \frac{k}{a}$ and$|G(j\omega)|dB \approx 20\log_{10}\left|\frac{k}{a}\right|$
- When ω is large, $|G(j\omega)| \approx \frac{k}{\omega}$ and$|G(j\omega)|dB \approx 20\log_{10}\left|\frac{k}{\omega}\right|$
- If $\omega = 10a$, then $|G(j10a)|dB \approx 20\log_{10}\left|\frac{k}{a}\right| - 20$
- If $\omega = 100a$, then $|G(j100a)|dB \approx 20\log_{10}\left|\frac{k}{a}\right| - 40$

Bode plot contd.,

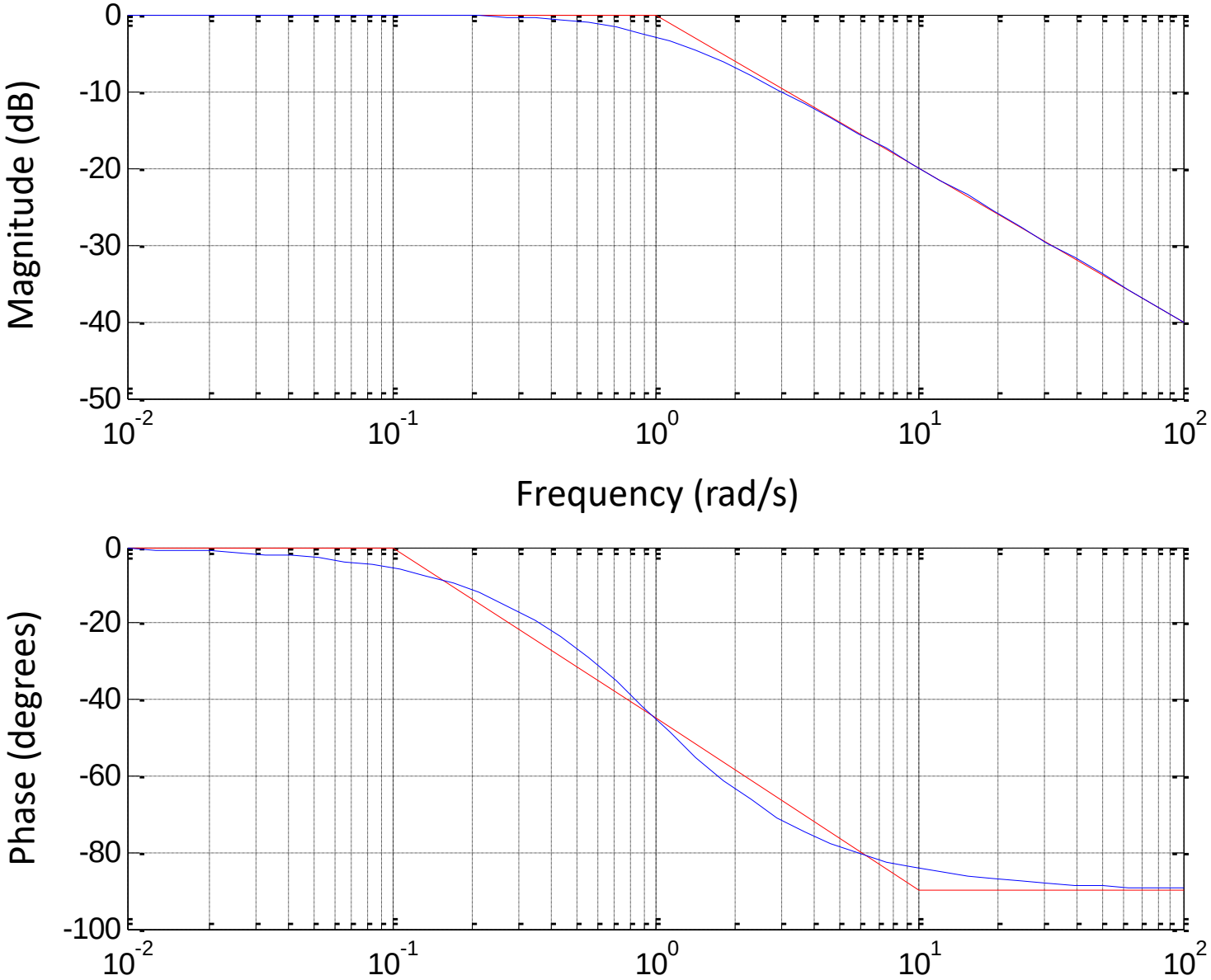
- On a log-log scale the gain follows the horizontal line $20\log_{10}\left|\frac{k}{a}\right|$ for small frequencies and then drops at 20dB per decade for large frequencies.
- It deviates most from these two asymptotes where they meet at $\omega = a$.
- This frequency is known as the break frequency.
- Here $|G(j\omega)| = \frac{k}{\sqrt{2a^2}}$,
- $|G(j\omega)|dB = 20\log_{10}\left|\frac{k}{a}\right| - 20\log_{10}\sqrt{2} = 20\log_{10}\left|\frac{k}{a}\right| - 3$

Bode plot contd.,

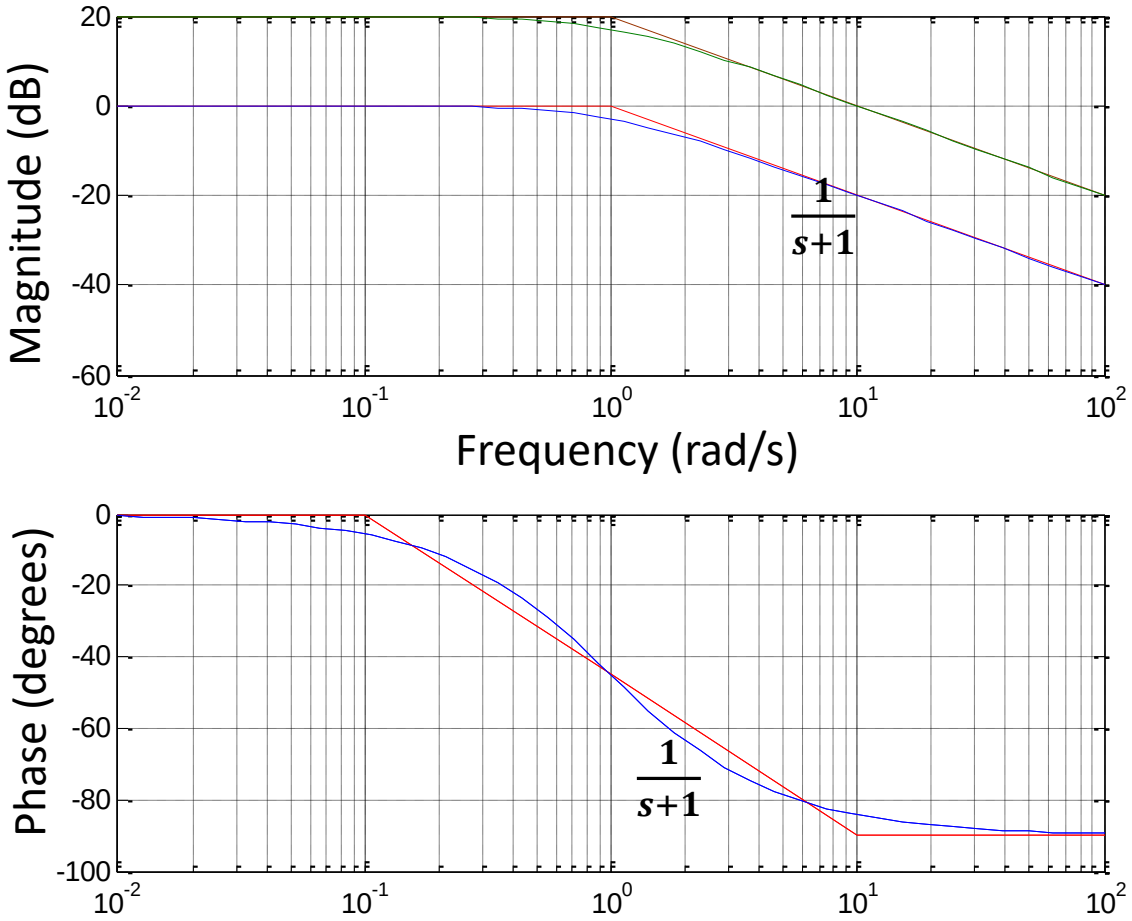
- On a Bode plot the phase is drawn on a linear scale, but against frequency on log scale.
- At $\omega = a$ we have $\angle G(ja) = \tan^{-1}\left(\frac{-a}{a}\right) = -\tan^{-1}\left(\frac{a}{a}\right) = -\frac{\pi}{4}$
- Near $\omega = a$ we may approximate $\angle G(j\omega) \approx -\frac{\pi}{4} - \frac{1}{2}\log\frac{\omega}{a}$
- This corresponds to a tangent that touches 0 when $\frac{\omega}{a} = e^{-\pi/2} \approx 0.2$ and touches $-\frac{\pi}{2}$ when $\frac{\omega}{a} = e^{\pi/2} \approx 0.5$.

Bode diagram of

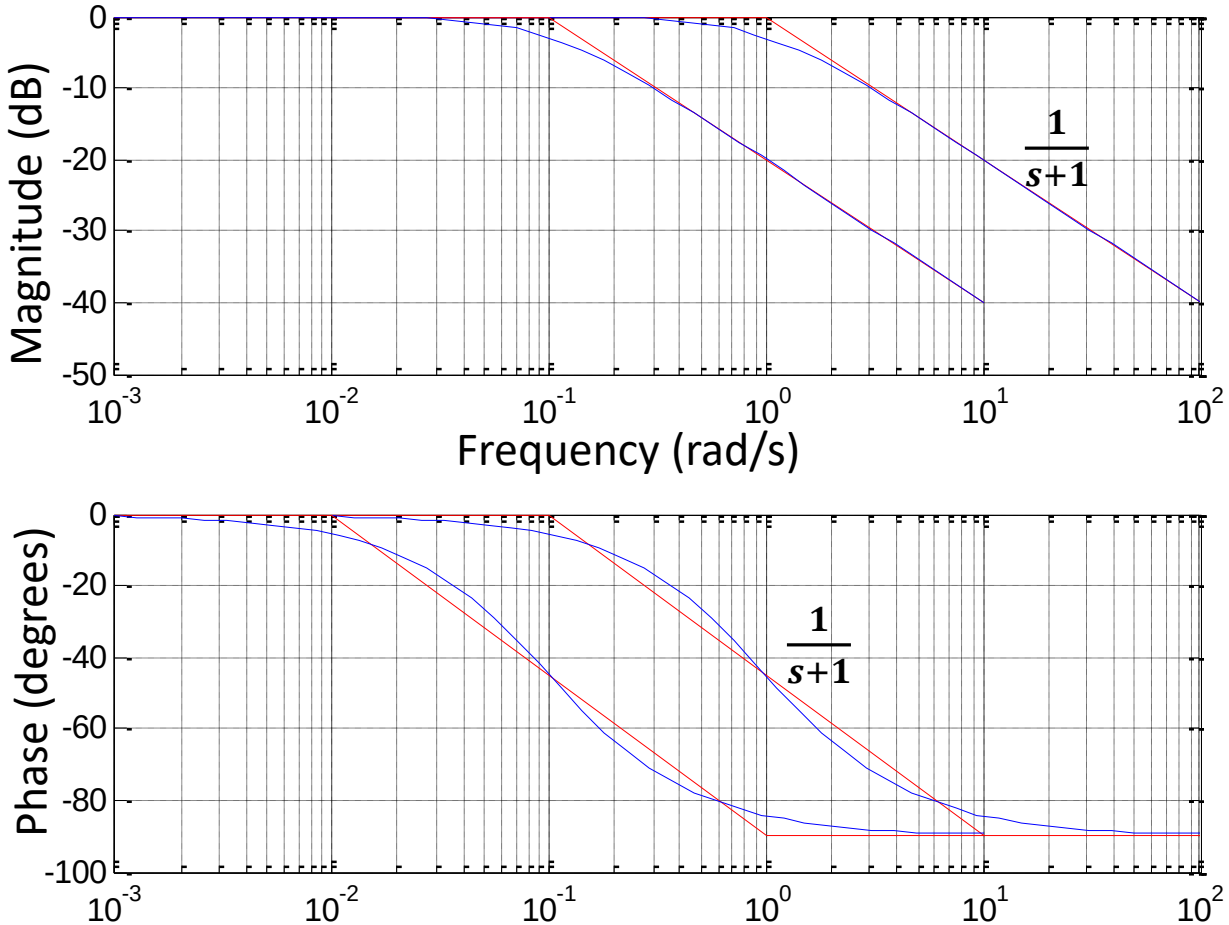
$$G(s) = \frac{1}{s+1}$$



Bode plot of $\frac{1}{s+1}$ vs. $\frac{10}{s+1}$



Bode plot of $\frac{1}{s+1}$ vs. $\frac{1}{10s+1}$



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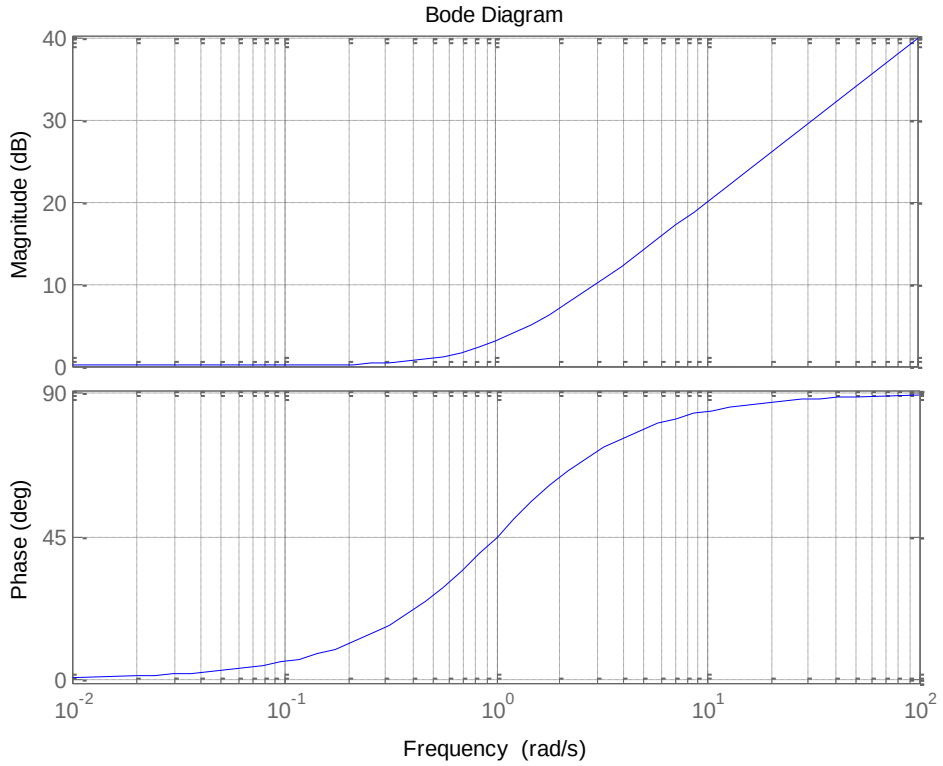
Analysis of 1st order elements: $G(s) = s + 1$

- $|(1 + j\omega)| = \sqrt{1 + \omega^2} \approx \begin{cases} 0\ dB\ for\ \omega \ll 1 \\ +3\ dB\ for\ \omega = 1 \\ 20\log_{10}\omega\ \ dB\ for\ \omega \gg 1 \end{cases}$

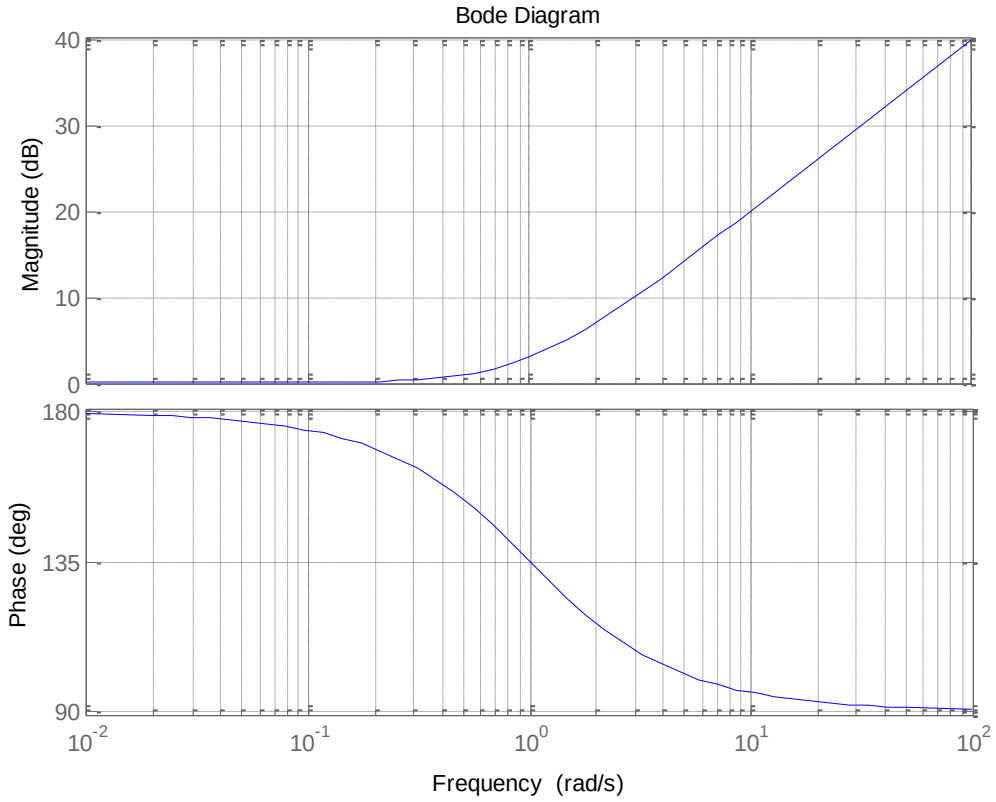
- $\angle(1 + j\omega) = \begin{cases} 0^o\ for\ \omega \to 0 \\ 45^o\ for\ \omega = 1 \\ 90^o\ for\ \omega \to \infty \end{cases}$



Analysis of 1st order elements: $G(s) = s - 1$

- $|(-1 + j\omega)| = \sqrt{1 + \omega^2} \approx \begin{cases} 0\ dB\ for\ \omega \ll 1 \\ +3\ dB\ for\ \omega = 1 \\ 20\log_{10}\omega\ \ dB\ for\ \omega \gg 1 \end{cases}$

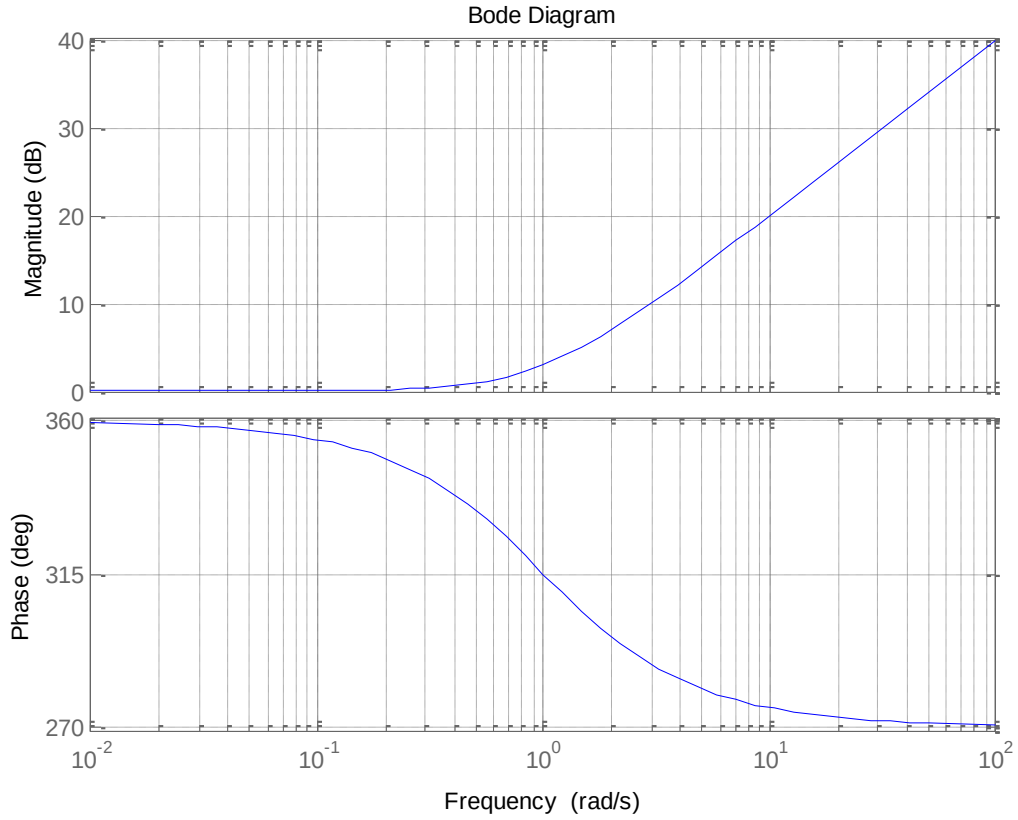
- $\angle(-1 + j\omega) = \begin{cases} -180^o\ for\ \omega \to 0 \\ -225^o\ for\ \omega = 1 \\ -270^o\ for\ \omega \to \infty \end{cases}$



Analysis of 1st order elements: $G(s) = 1 - s$

- $|(1 - j\omega)| = \sqrt{1 + \omega^2} \approx \begin{cases} 0\ dB\ for\ \omega \ll 1 \\ +3\ dB\ for\ \omega = 1 \\ 20\log_{10}\omega\ \ dB\ for\ \omega \gg 1 \end{cases}$

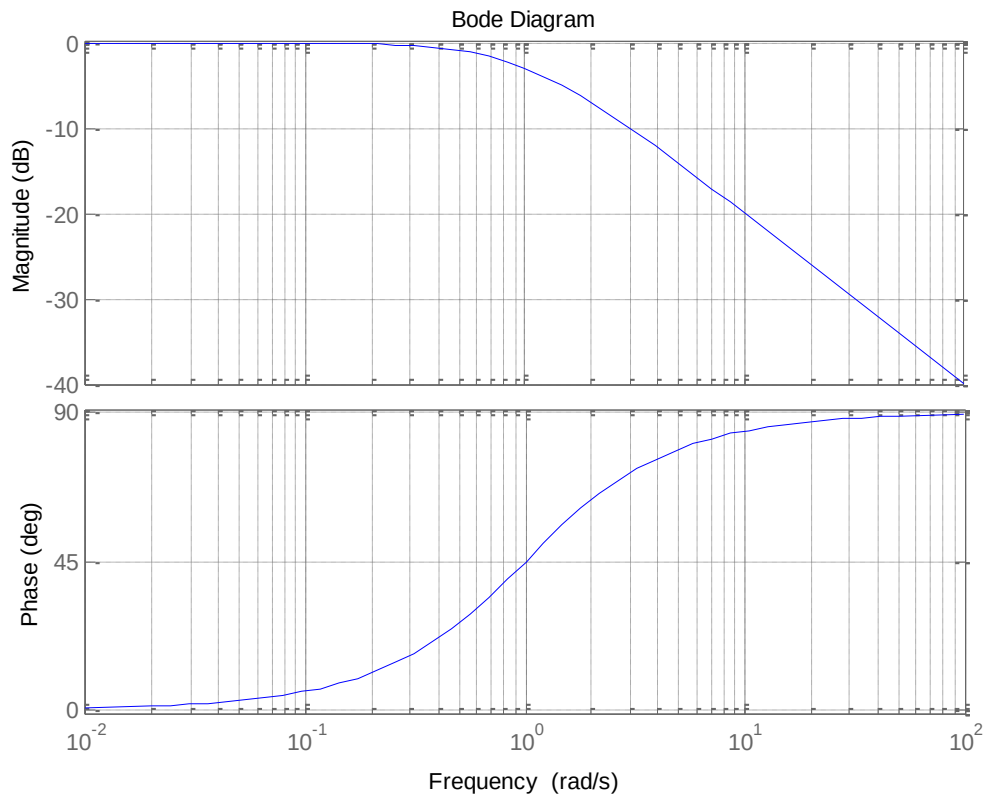
- $\angle(1 - j\omega) = \begin{cases} 0^o\ for\ \omega \to 0 \\ -45^o\ for\ \omega = 1 \\ -90^o\ for\ \omega \to \infty \end{cases}$



Analysis of 1st order elements: $G(s) = \frac{1}{1-s}$

- $\left|\frac{1}{(1-j\omega)}\right| = \frac{1}{\sqrt{1+\omega^2}} \approx \begin{cases} 0\ dB\ for\ \omega \ll 1 \\ -3\ dB\ for\ \omega = 1 \\ 20\log_{10}\omega\ \ dB\ for\ \omega \gg 1 \end{cases}$

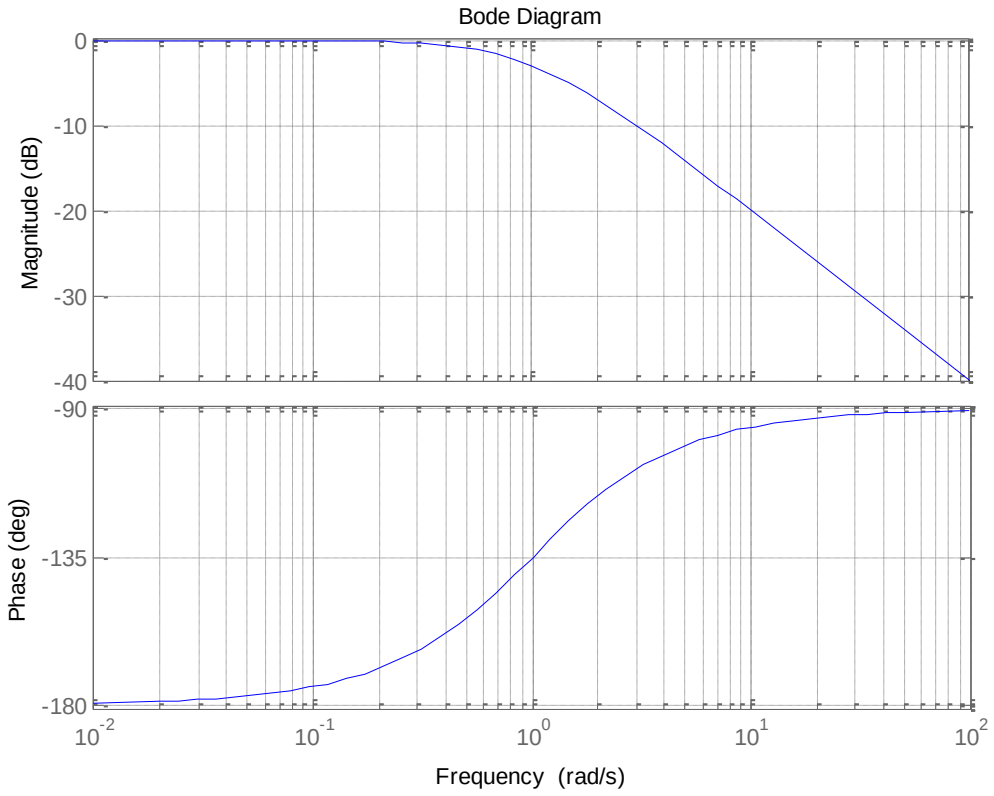
- $\angle(1-j\omega) = \begin{cases} -360^o\ for\ \omega \to 0 \\ -315^o\ for\ \omega = 1 \\ -270^o\ for\ \omega \to \infty \end{cases}$



Analysis of 1st order elements: $G(s) = \frac{1}{s-1}$

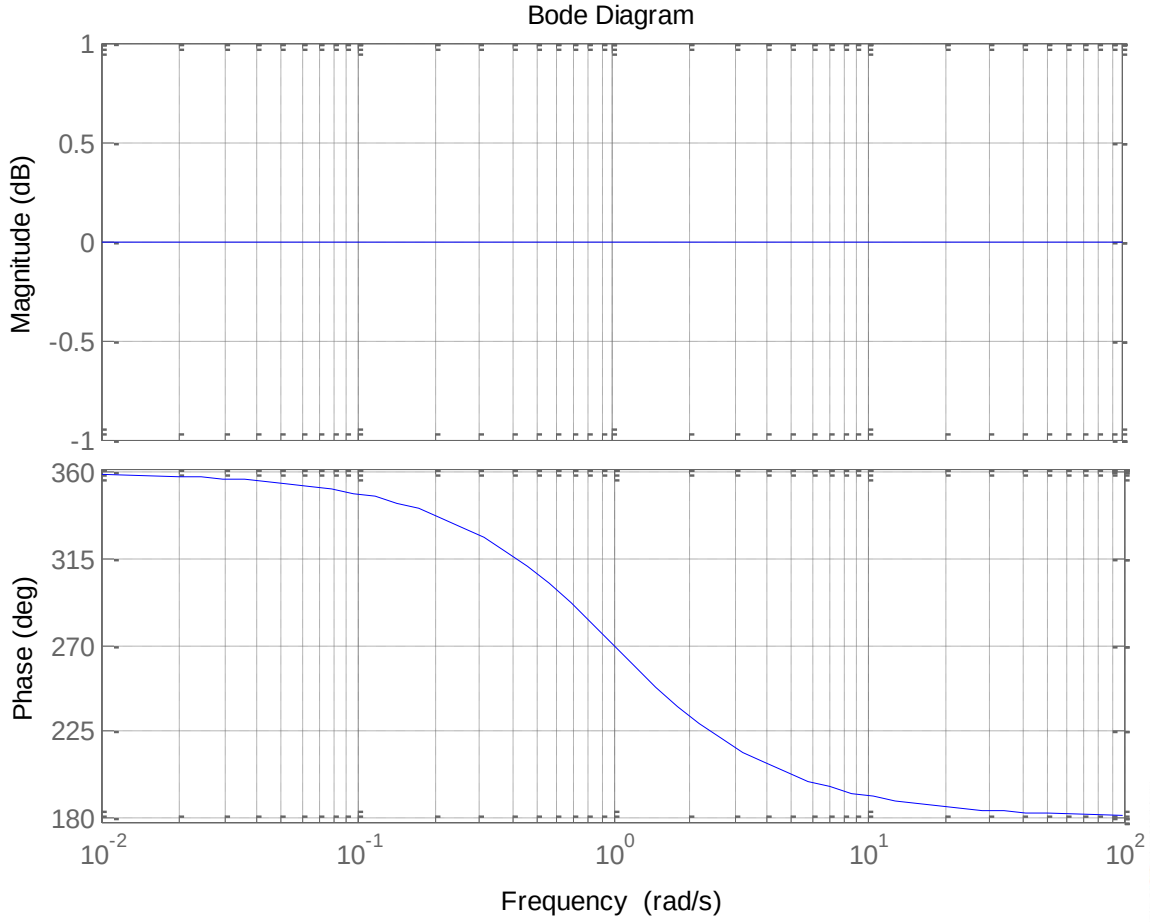
- $\left|\frac{1}{(j\omega-1)}\right| = \frac{1}{\sqrt{1+\omega^2}} \approx \begin{cases} 0\ dB\ for\ \omega \ll 1 \\ -3\ dB\ for\ \omega = 1 \\ 20\log_{10}\omega\ \ dB\ for\ \omega \gg 1 \end{cases}$

- $\angle(j\omega - 1) = \begin{cases} -180^o\ for\ \omega \to 0 \\ -135^o\ for\ \omega = 1 \\ -90^o\ for\ \omega \to \infty \end{cases}$



First order system $G(s) = \frac{1-s}{1+s}$

- $|G(j\omega)|dB = 20\log_{10}\sqrt{1+\omega^2} - 20\log_{10}\sqrt{1+\omega^2} = 0$
- $\angle G(j\omega) = \angle(1-j\omega) - \angle(1+j\omega) = -2\angle(1+j\omega)$



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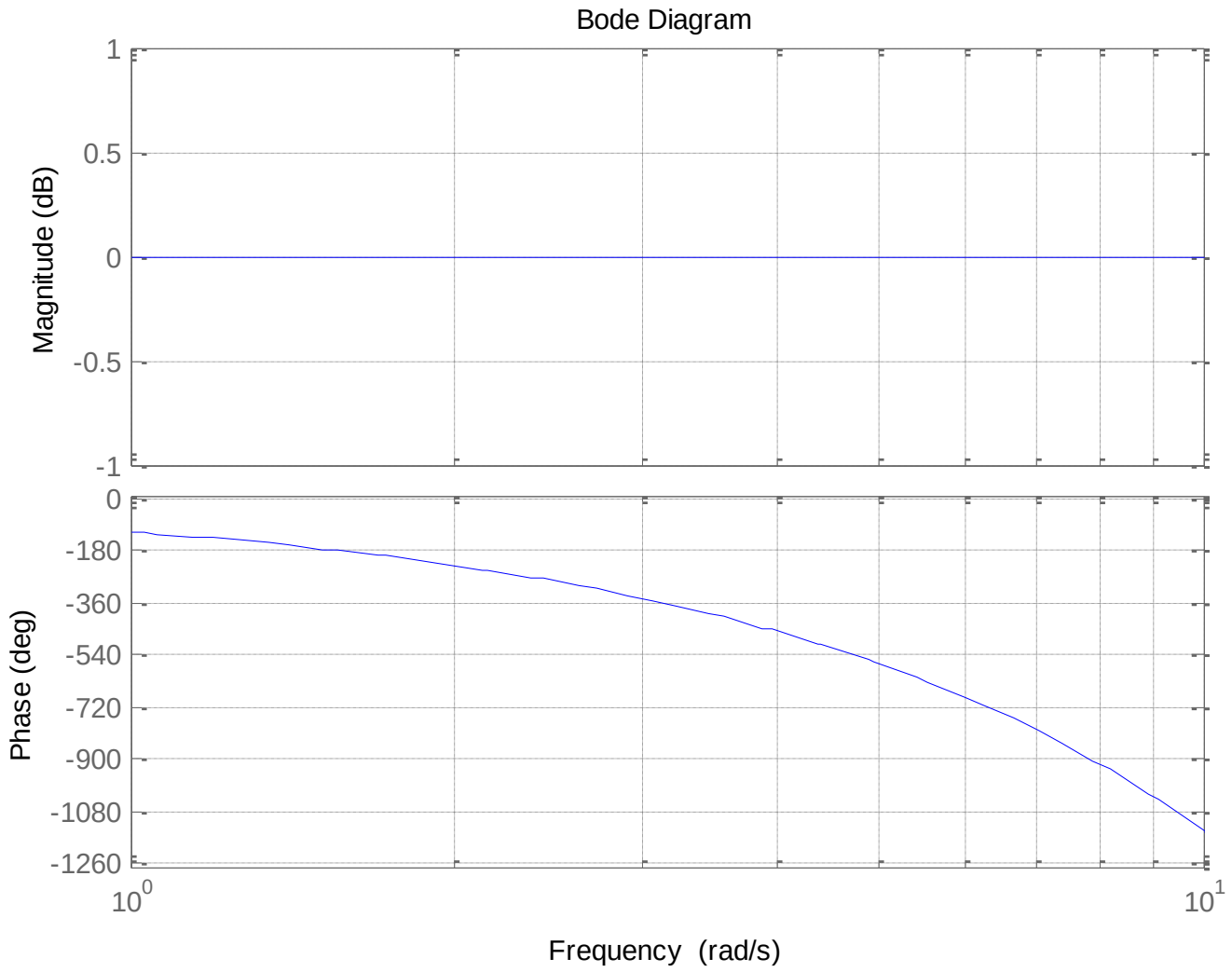
Input delay model: $G(s) = e^{-\tau s}$

- Time delay = τ
- $G(j\omega) = 1, G(j\omega)dB = 0$
- $\angle G(j\omega) = -\omega\tau$
- Bode plot of $G(s) = e^{-2s}$



Second order systems

- Consider the generic second order system with steady state gain *k:*

$$G(s) = \frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(j\omega) = \frac{k\omega_n^2}{-\omega^2 + j2\zeta\omega_n\omega + \omega_n^2} = \frac{k}{\left(1 - \frac{\omega^2}{\omega_n^2}\right) + j\,\frac{2\zeta\omega}{\omega_n}}$$

- When ω is small $G(j\omega) \approx k, |G(j\omega)|dB \approx 20\log_{10} k\,, \angle G(j\omega) \approx 0^o$
- When ω is large

$$G(j\omega) \approx -\frac{k\omega_n^2}{\omega^2}, |G(j\omega)|dB \approx 20\log_{10} k + 40\log_{10}\omega_n - 40\log_{10}\omega$$

$$\angle G(j\omega) \approx -180^o$$

- When $\omega = \omega_n$, $G(j\omega) = \frac{k}{j2\zeta}, |G(j\omega)|dB \approx 20\log_{10} k - 20\log_{10} 2\zeta$

$$\angle G(j\omega) \approx -90^o$$

Second order systems contd.,

- If the damping ratio $\zeta < {}^{1}/_{\sqrt{2}}$, there is a peak in the gain.
- Since $G(j\omega) = \frac{k\omega_n^2}{\omega_n^2 - \omega^2 + j2\zeta\omega_n\omega}, |G(j\omega)|^2 = \frac{\left(k\omega_n^2\right)^2}{\left(\omega_n^2 - \omega^2\right)^2 + (2\zeta\omega_n\omega)^2}$
- The peak in $|G(j\omega)|$ occurs where $f(\omega^2) = (\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2$ is minimum.
- Differentiating $f(\omega^2)$ with respect to ω^2 gives

$$\frac{df(\omega^2)}{d\omega^2} = -2(\omega_n^2 - \omega^2) + 4\zeta^2\omega_n^2$$

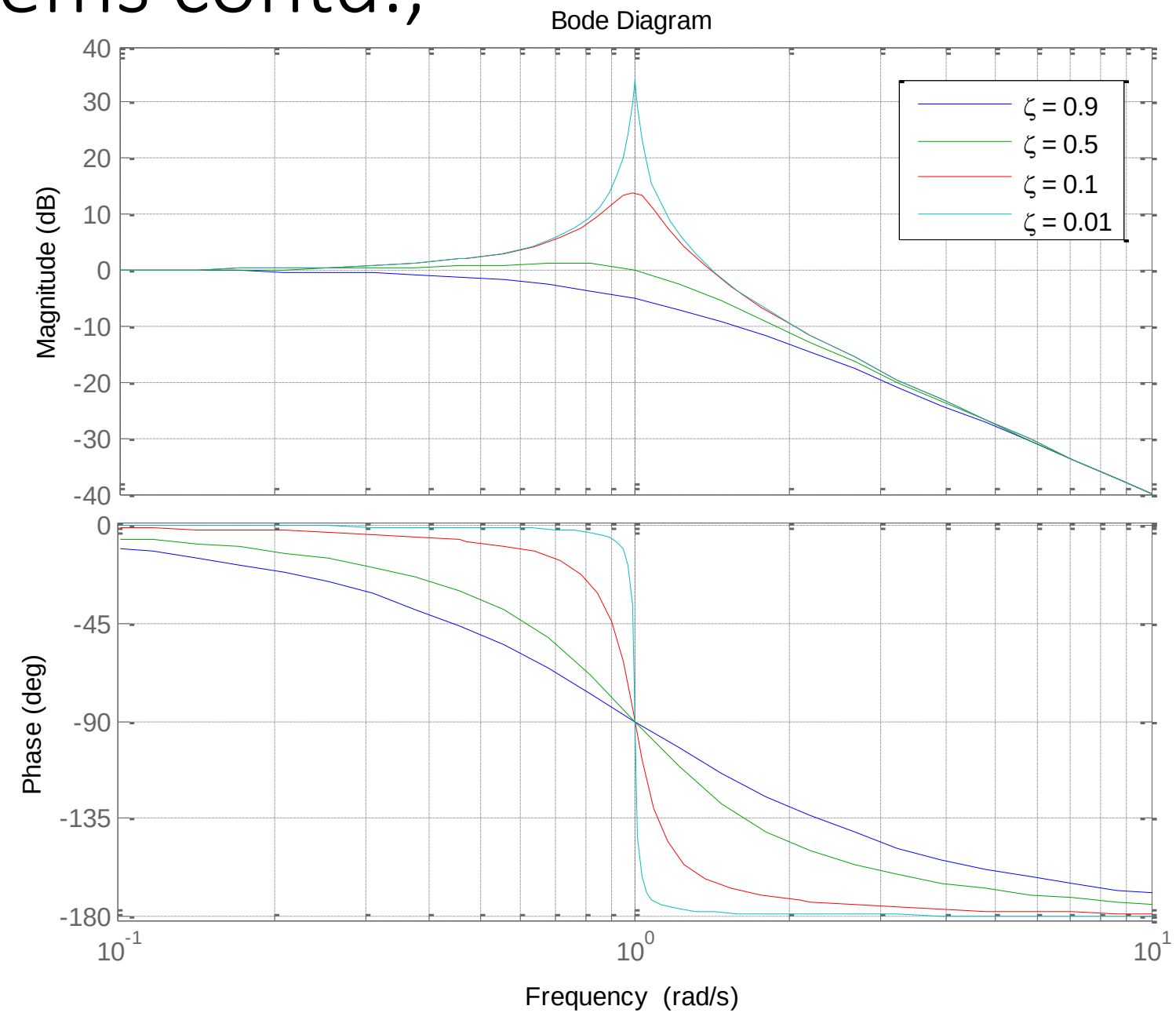
which becomes 0 when $\omega^2 = \omega_n^2(1 - 2\zeta^2), i.e., \omega = \omega_n\sqrt{(1 - 2\zeta^2)}$

- At this frequency $G(j\omega)_{max} = \frac{k}{2\zeta^2 + j2\zeta\sqrt{(1-2\zeta^2)}}, |G(j\omega)|_{max} = \frac{k}{2\zeta\sqrt{(1-\zeta^2)}}$

Second order systems contd.,

$$|G(j\omega)|_{max} = \frac{k}{2\zeta\sqrt{(1-\zeta^2)}}$$

As $\zeta \rightarrow 0, |G(j\omega)|_{max} \rightarrow \infty$



Derivation of the main result

- Consider the stable system $G(s) = \frac{b(s)}{(s+p_1)(s+p_2)(s+p_3)\cdots(s+p_n)}$
- Suppose that the input is $u(t) = \sin \omega t$, $\mathcal{L}u(t) = U(s) = \frac{\omega}{s^2+\omega^2}$
- The output is $Y(s) = G(s)U(s)$
- Hence $Y(s) = \frac{b(s)}{(s+p_1)(s+p_2)(s+p_3)\cdots(s+p_n)} \frac{\omega}{(s-j\omega)(s+j\omega)}$
- $Y(s) = \frac{A_1}{(s+j\omega)} + \frac{A_2}{(s-j\omega)} + \frac{B_1}{(s+p_1)} + \frac{B_2}{(s+p_2)} + \frac{B_3}{(s+p_3)} \cdots + \frac{B_n}{(s+p_n)}$
- Therefore

$$y(t) = A_1 e^{-j\omega t} + A_2 e^{-j\omega t} + B_1 e^{-p_1 t} + B_2 e^{-p_2 t} + B_3 e^{-p_3 t} \cdots + B_n e^{-p_n t}$$

- At steady state

$$y_{ss}(t) = A_1 e^{-j\omega t} + A_2 e^{-j\omega t}$$

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Derivation of the main result contd.,

- From the partial fraction expansion, we find that
- $A_1 = G(s)\frac{\omega}{s-j\omega}\Big|_{s=-j\omega} = j\frac{1}{2}G(-j\omega)$
- $A_2 = G(s)\frac{\omega}{s+j\omega}\Big|_{s=j\omega} = -j\frac{1}{2}G(j\omega)$
- It is possible to show that

$$|G(-j\omega)| = |G(j\omega)| \text{ and } \angle G(-j\omega) = -\angle G(j\omega)$$

- Hence $y_{ss}(t) = A_1 e^{-j\omega t} + A_2 e^{-j\omega t} = j\frac{1}{2}G(-j\omega)e^{-j\omega t} - j\frac{1}{2}G(j\omega)e^{-j\omega t}$

$$y_{ss}(t) = j\frac{1}{2}|G(j\omega)|e^{-j\angle G(j\omega)}e^{-j\omega t} - j\frac{1}{2}|G(j\omega)|e^{j\angle G(j\omega)}e^{j\omega t}$$

$$y_{ss}(t) = j\frac{1}{2}|G(j\omega)| \times (-2j)\sin\big(\omega t + \angle G(j\omega)\big)$$

$$y_{ss}(t) = |G(j\omega)|\sin\big(\omega t + \angle G(j\omega)\big)$$

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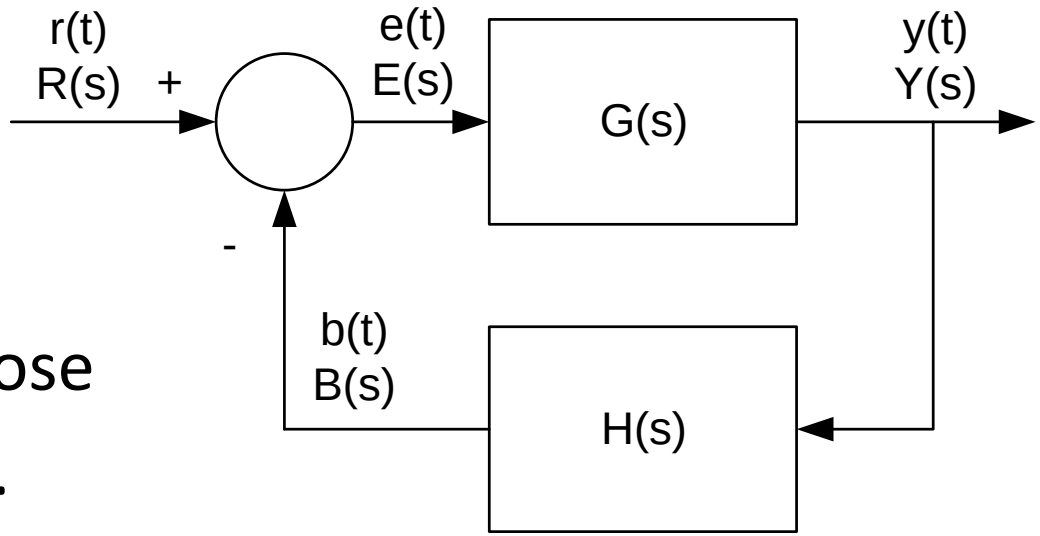
Bode diagram applications

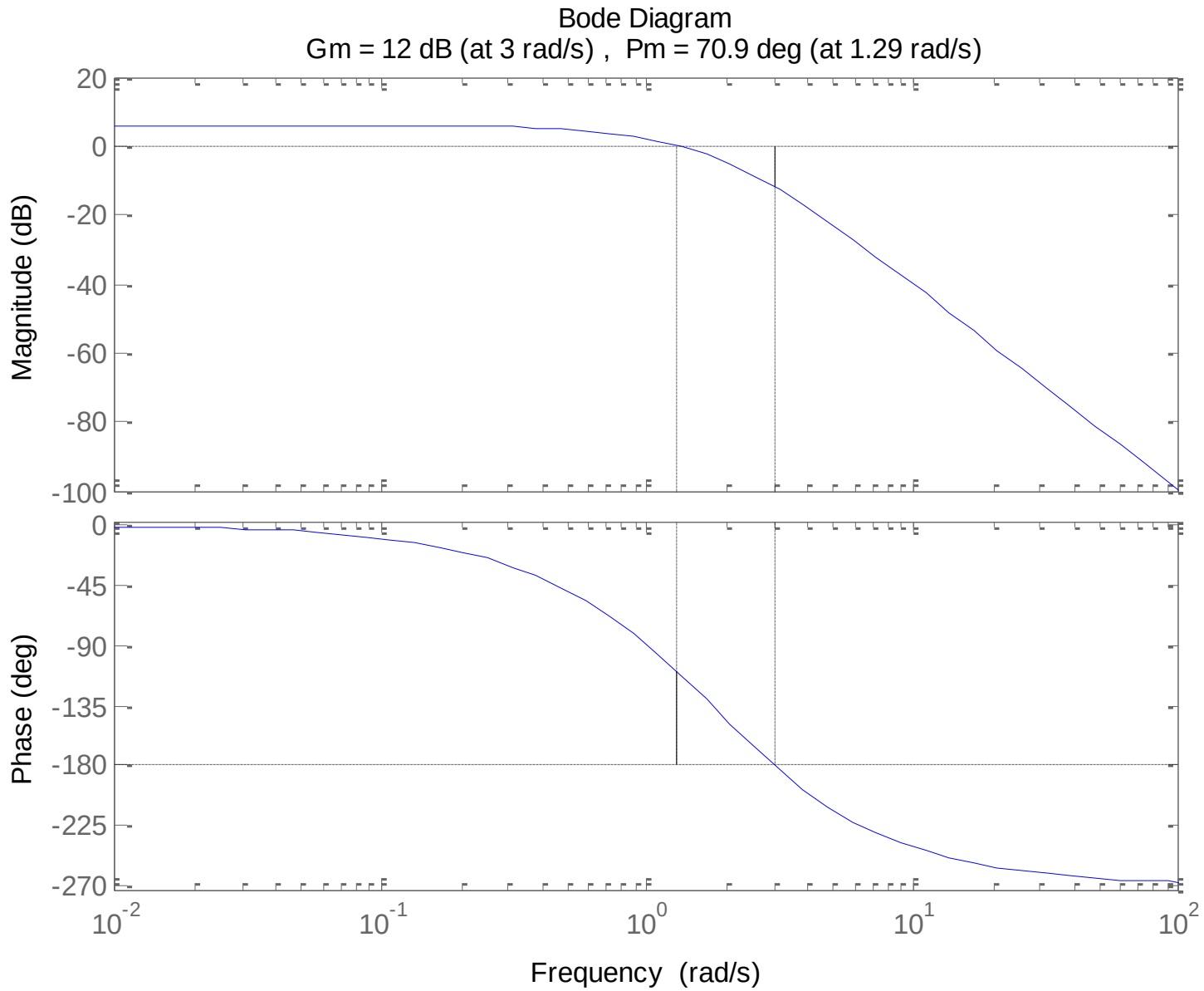


- Consider the feedback control system whose closed loop transfer function is $\frac{G(s)}{1+G(s)H(s)}$.
- It can be seen that stability is lost at
- $1 + G(j\omega)H(j\omega) = 0$, i.e., $G(j\omega)H(j\omega) = -1$
- That is $|G(j\omega)H(j\omega)| = 1$ i.e., $|G(j\omega)H(j\omega)|dB = 0$and $\angle G(j\omega)H(j\omega) = -180^0$
- From the Bode diagram of $G(j\omega)H(j\omega)$, the Margin to loose stability can be found graphically

Bode diagram applications contd.,

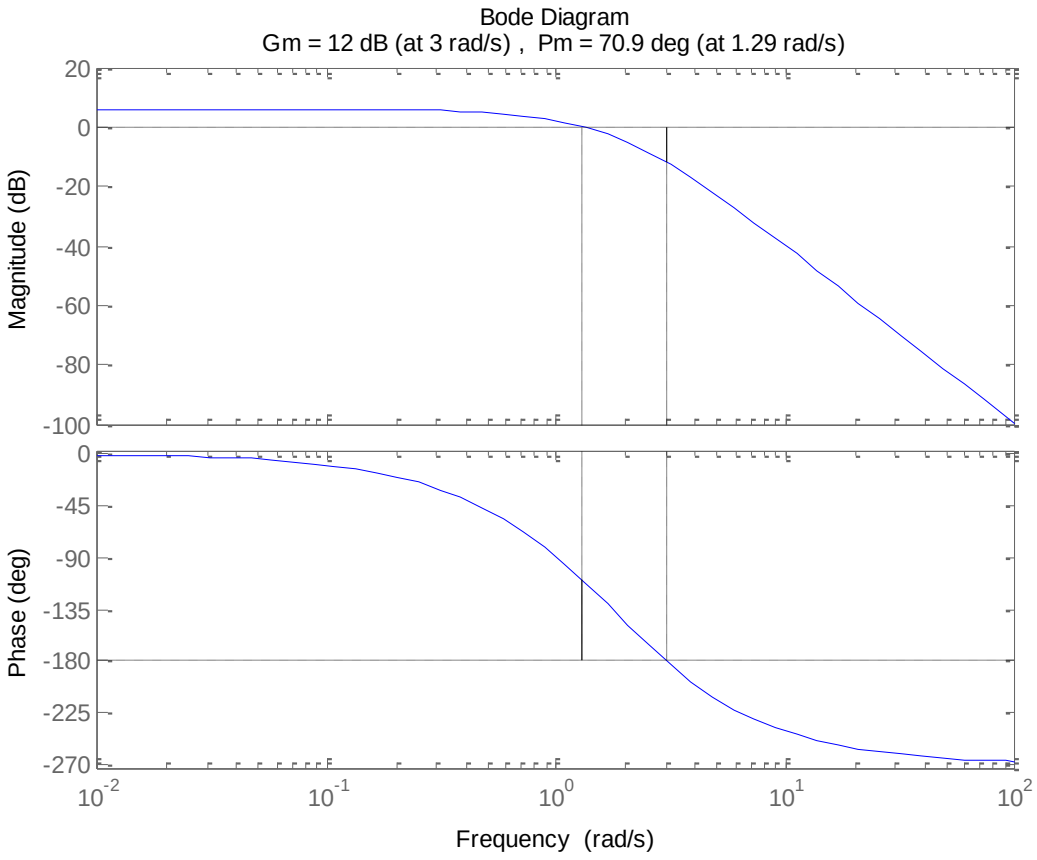
- Consider

$$G(s)H(s) = \frac{10}{s^3+5s^2+9s+5}$$



Bode diagram applications contd.,

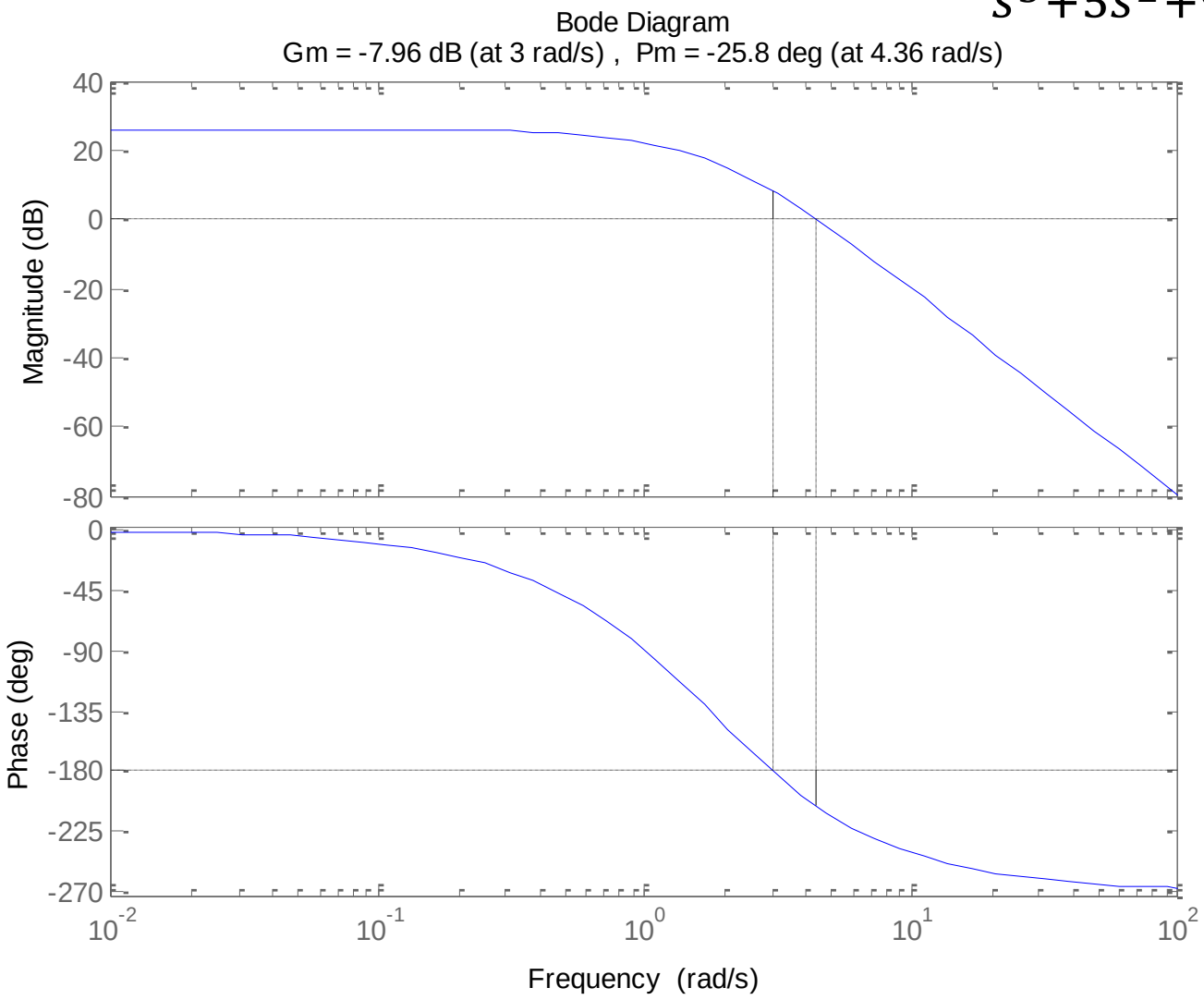
- Gain margin = 12 dB
- Phase crossover frequency = 3 rad/s
- Phase margin = 70.9 degrees
- Gain crossover frequency = 1.29 rad/s



Bode diagram applications contd.,

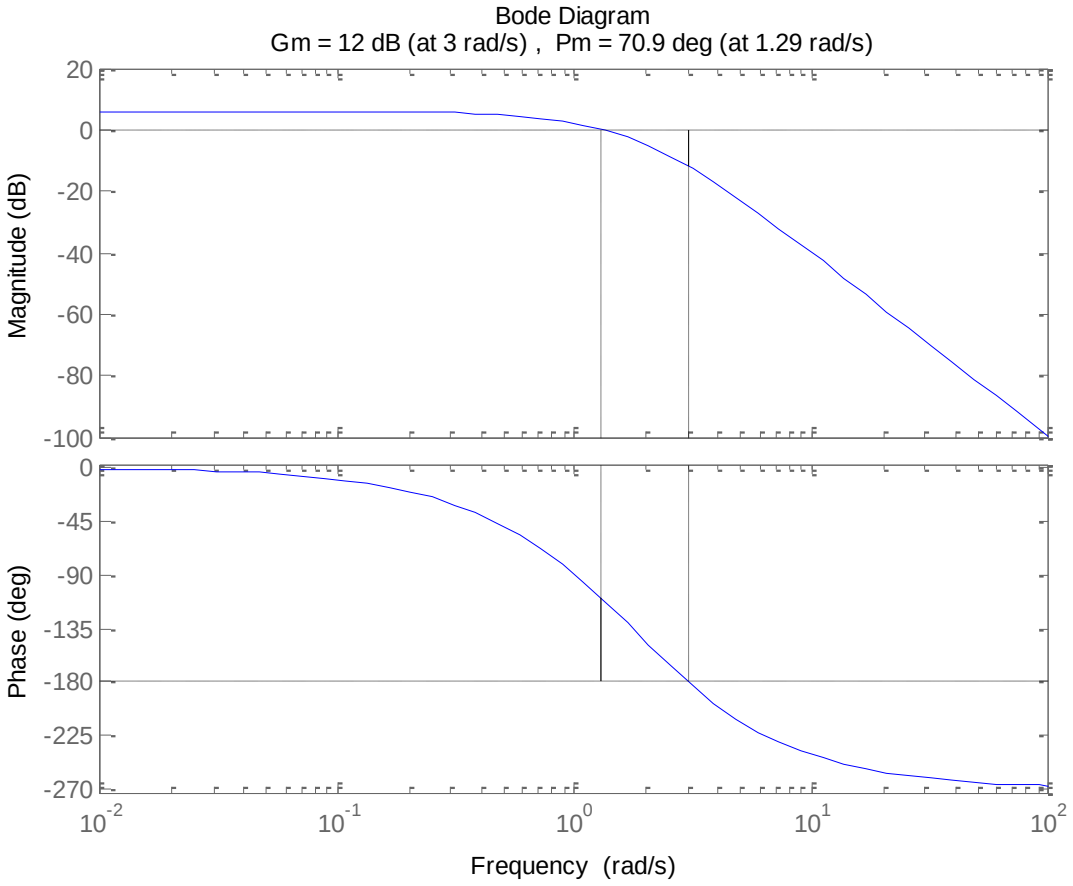
- Consider the same system with higher gain $G(s)H(s) = \frac{100}{s^3+5s^2+9s+5}$

Change in the **gain** changes the gain margin. But keeps the **Phase crossover frequency unchanged**. However, it changes both the phase margin and the gain crossover frequency.

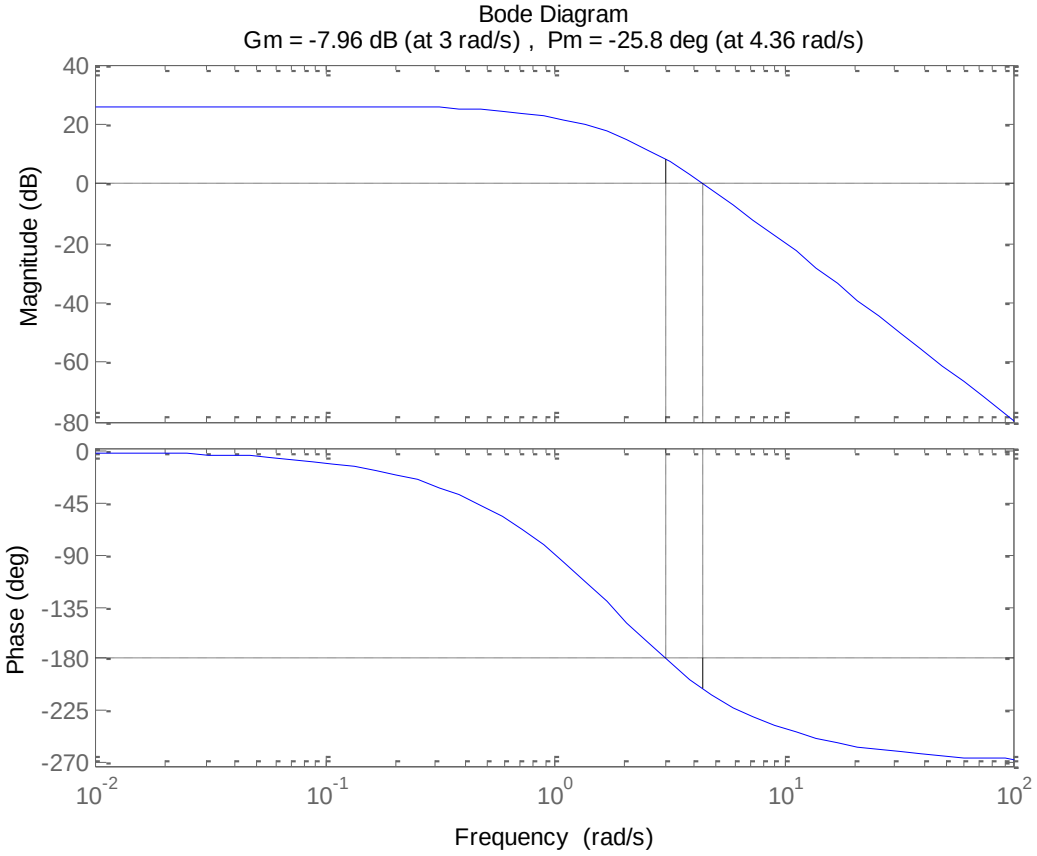


Bode diagram applications contd.,

Original system

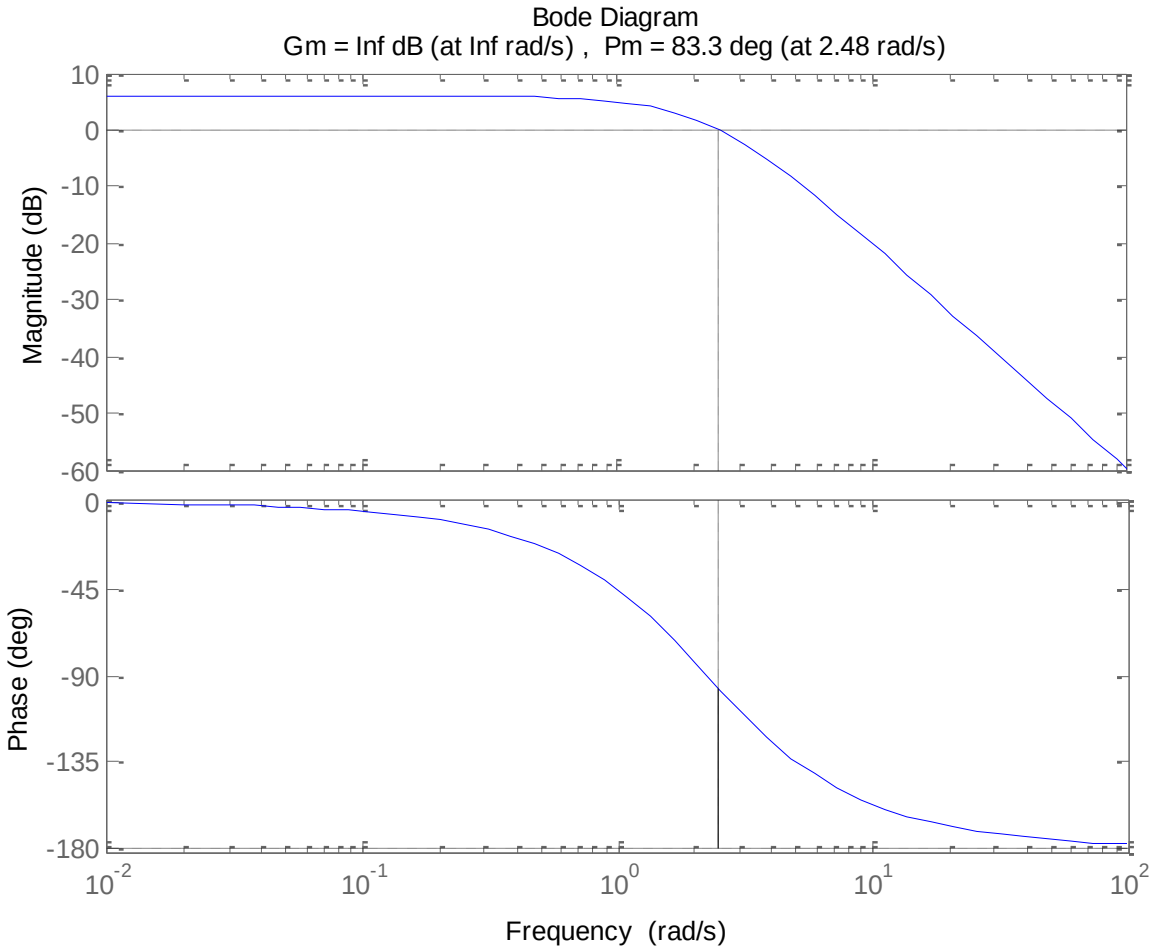


System with increased gain



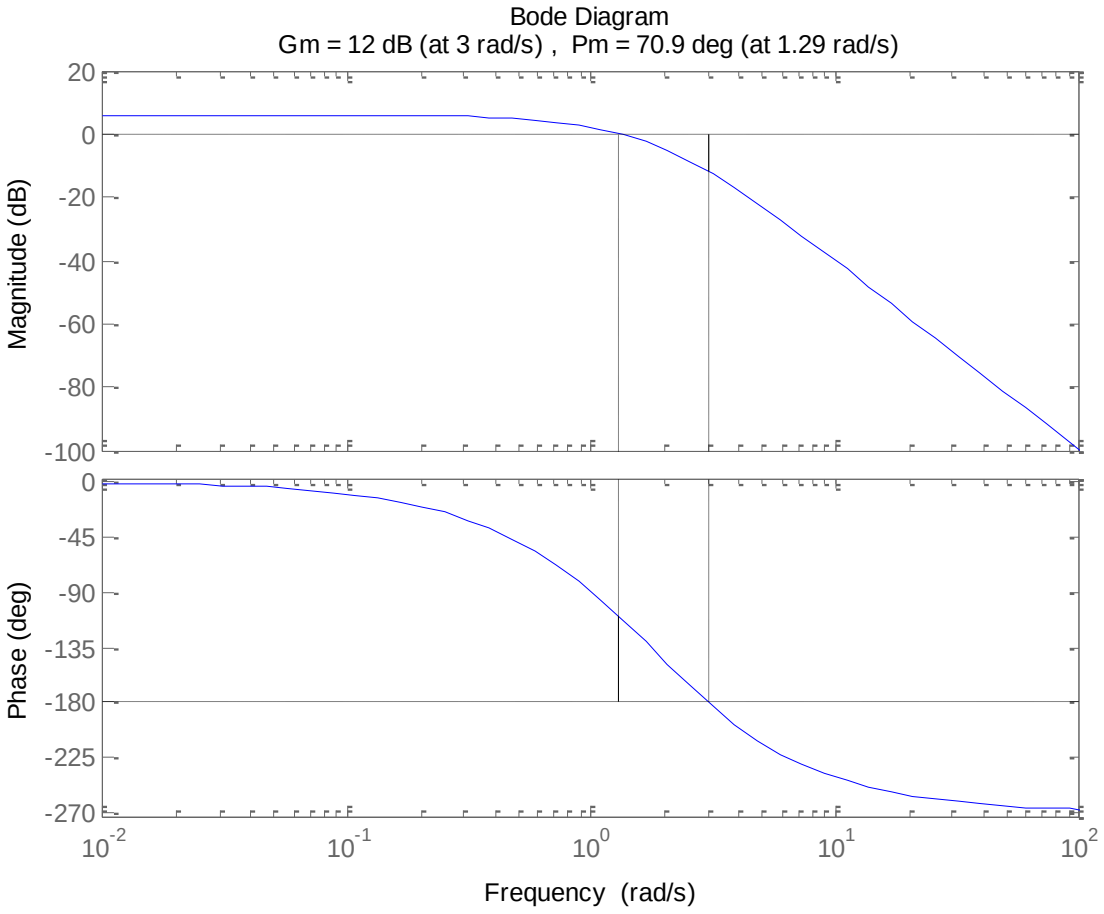
Bode diagram applications contd.,

- Consider adding a zero to the original system such that $G(s)H(s) = \frac{10s+10}{s^3+5s^2+9s+5}$
- It has increased the phase margin

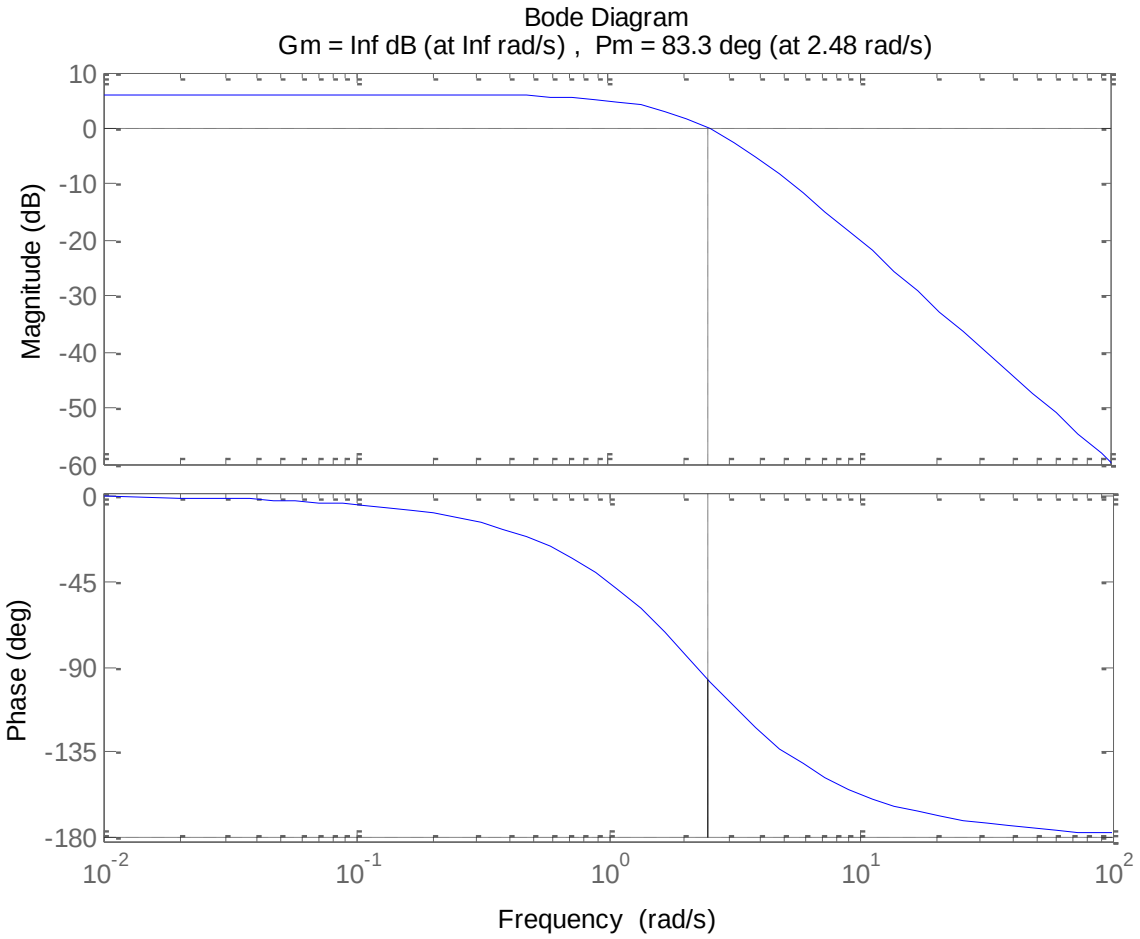


Bode diagram applications contd.,

Original system



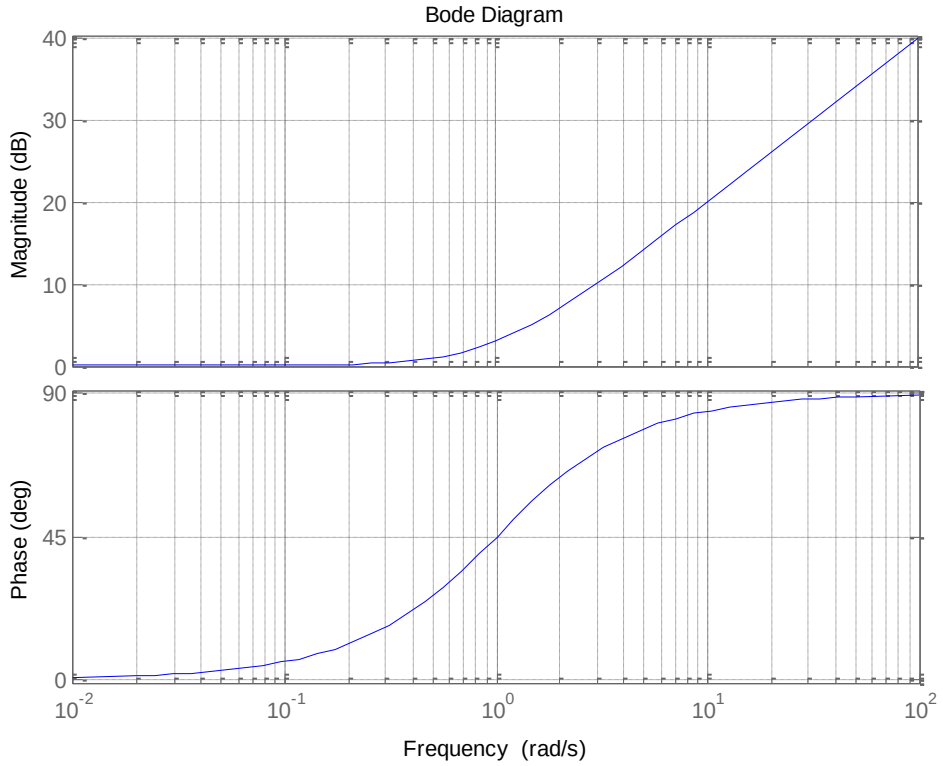
System with added zero at s = -1



Recall the function of a Zero $G(s) = s + 1$

- $|(1 + j\omega)| = \sqrt{1 + \omega^2} \approx \begin{cases} 0\ dB\ for\ \omega \ll 1 \\ +3\ dB\ for\ \omega = 1 \\ 20\log_{10}\omega\ \ dB\ for\ \omega \gg 1 \end{cases}$

- $\angle(1 + j\omega) = \begin{cases} 0^o\ for\ \omega \to 0 \\ 45^o\ for\ \omega = 1 \\ 90^o\ for\ \omega \to \infty \end{cases}$



Bode diagram applications contd.,

- High gain margin is preferred
- Phase margin is typically maintained between 35 to 70 degrees